Codex Future Technologies

SERVICE FEE ANALYSIS

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□ Service Fee Analysis Project — Data Analytics Report

📚 Project Overview

This project analyses **Service Fee data** across multiple operational areas — from transactions and procedures to financial performance, provider workload, and patient demographics.
The objective is to identify **revenue opportunities**, **operational bottlenecks**, and **performance insights** that help management make **data-driven decisions**.

The dataset was analysed and visualized using **Power BI**, and dashboards were created for each functional domain:

1. **Transaction Dashboard**
2. **Procedure Dashboard**
3. **Units and Workload Dashboard**
4. **Finance Dashboard**
5. **Provider and Operational Dashboard**
6. **Patient and Demographics Dashboard**

Each dashboard includes KPIs, slicers for interactive filtering, and page navigation for smooth report exploration.

□ 1. Transaction Dashboard

🎯 Purpose

To track payment inflows, collection efficiency, and billing trends across different payers and service locations.

📊 Key Visuals

- **Revenue by Primary Payer:** Clustered column chart showing contribution of each payer to total revenue.
- **Payment Collection Percentage:** Gauge chart displaying efficiency of collections vs billed amount.
- **Daily Case Volume Analysis:** Column chart monitoring operational case trends per day.
- **Billing Distribution by Place of Service:** Pie chart visualizing billing patterns across service locations.
- **Net Payment & Actual Allowed by Month/Year:** Line chart showing long-term payment trends.

📌 KPIs

- **Doctor Income**
- **Average Processing Days per Doctor**
- **Contractual Adjustment (Total)**
- **Outstanding Balance**

🔍 Slicers

- Place of Service
- Primary Payer
- Date of Service

🔔 Managerial Insights

- Identifies top-performing payers driving maximum revenue.
- Reveals delay trends through processing time KPIs.
- Helps track outstanding payments and uncollected revenue.
- Seasonal case patterns support workload and staffing decisions.

01.TRANSACTION DASHBOARD :-

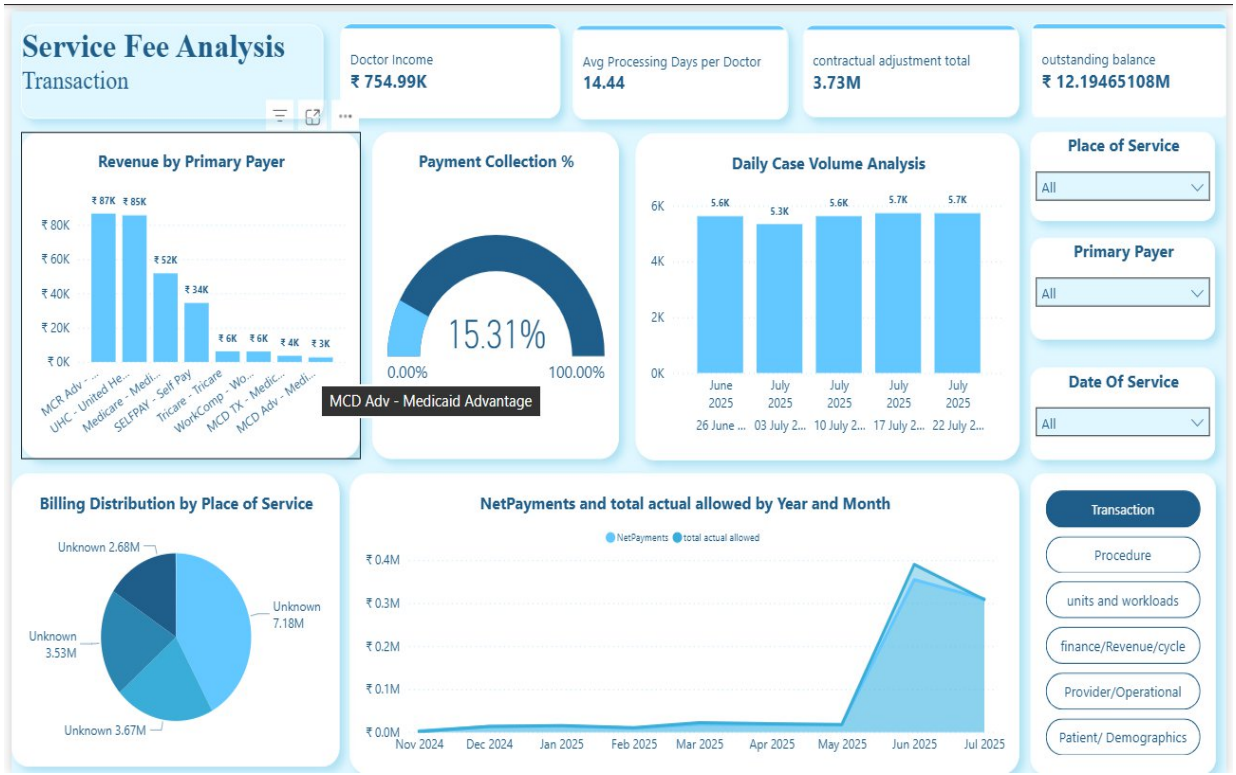

FIGURE .1 TRANSACTION DASHBOARD

⚙ 2. Procedure Dashboard

🎯 Purpose

To assess profitability and performance of medical procedures, CPT codes, and ASA classifications.

📊 Key Visuals

- **Net Payment by CPT Code:** Clustered bar chart showing most profitable procedures.
- **Expected vs Received Collection:** Gauge chart to monitor financial accuracy and recovery rate.
- **Revenue by CPT Description:** Row chart identifying high-revenue procedures.
- **Net Payment by ASA Description:** Pie chart showing case mix by anesthesia type.
- **Complexity Trend by RVU & Service Category:** Line chart tracking workload trends.

📌 KPIs

- Net Payment
- Utilization Rate
- Total Revenue vs Total Procedures

🔍 Slicers

- Date of Service
- CPT Code
- ASA Description

🔔 Managerial Insights

- Reveals which procedures generate the highest financial return.
- Highlights underperforming codes or categories for review.
- Enables optimization of case mix and resource allocation.

2.Procedure dashboard

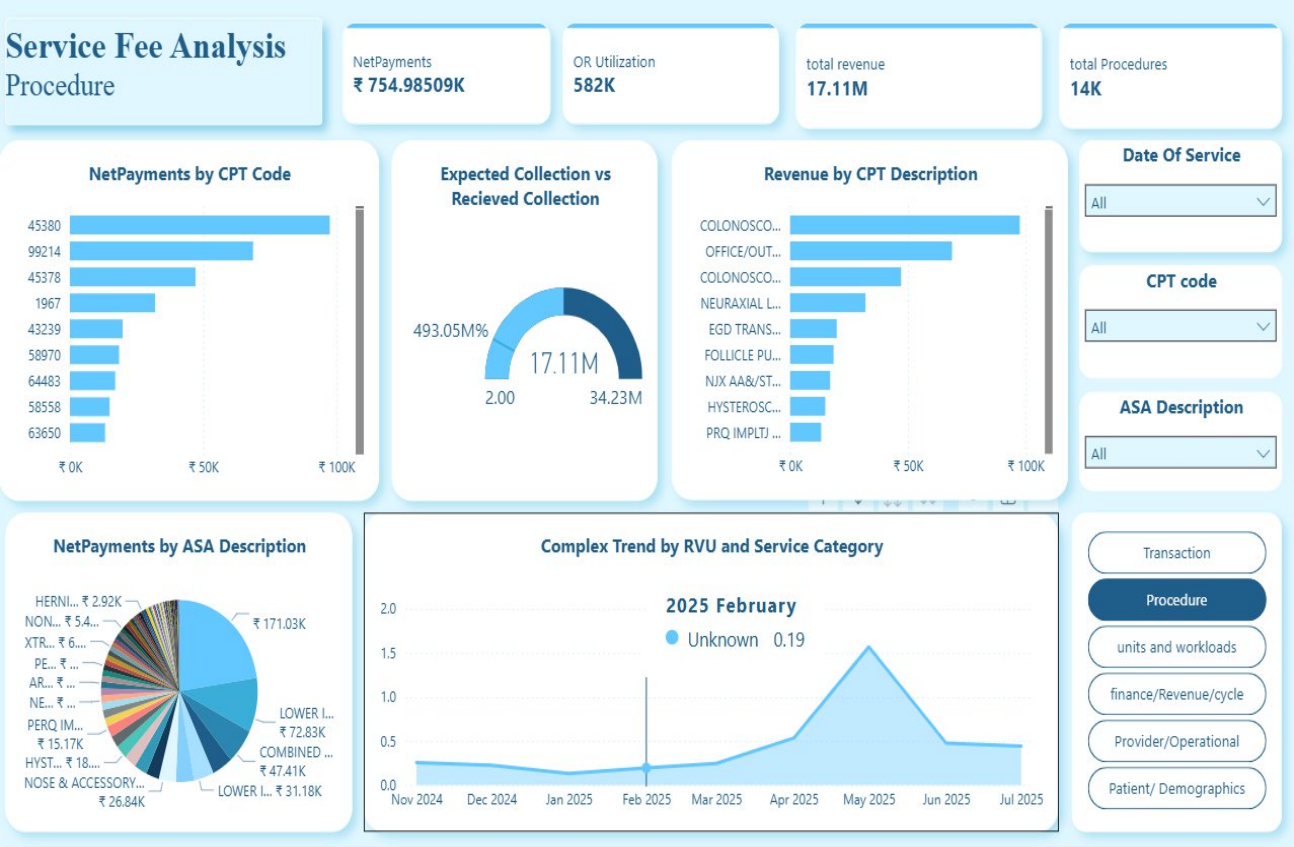

Figure .02 PROCEDURE DASHBOARD

3. Units & Workload Dashboard

🎯 Purpose

To monitor productivity, efficiency, and workload distribution among providers.

📊 Key Visuals

- **Sum of Work RVU by Rendering Provider AQI ID:** Clustered bar chart showing individual productivity.
- **Distribution of Procedure Complexity:** Gauge chart highlighting complexity mix.
- **Revenue by CPT Description:** Column chart linking financial outcomes to specific procedures.
- **Net Payment by ASA Description:** Category-level revenue view.
- **Complexity Trend by RVU & Service Category:** Line chart showing workload variation.

📌 KPIs

- Revenue per RVU
- Average Time per Case
- Average Work RVU per Case
- Total Anesthesia Units

🔍 Slicers

- Date of Service
- CPT Code
- ASA RVU

💡 Managerial Insights

- Tracks which providers handle higher complexity cases efficiently.
- Helps balance workloads and improve utilization.
- Monitors unit-based productivity for fair incentive planning.

3.Units and Workload :-

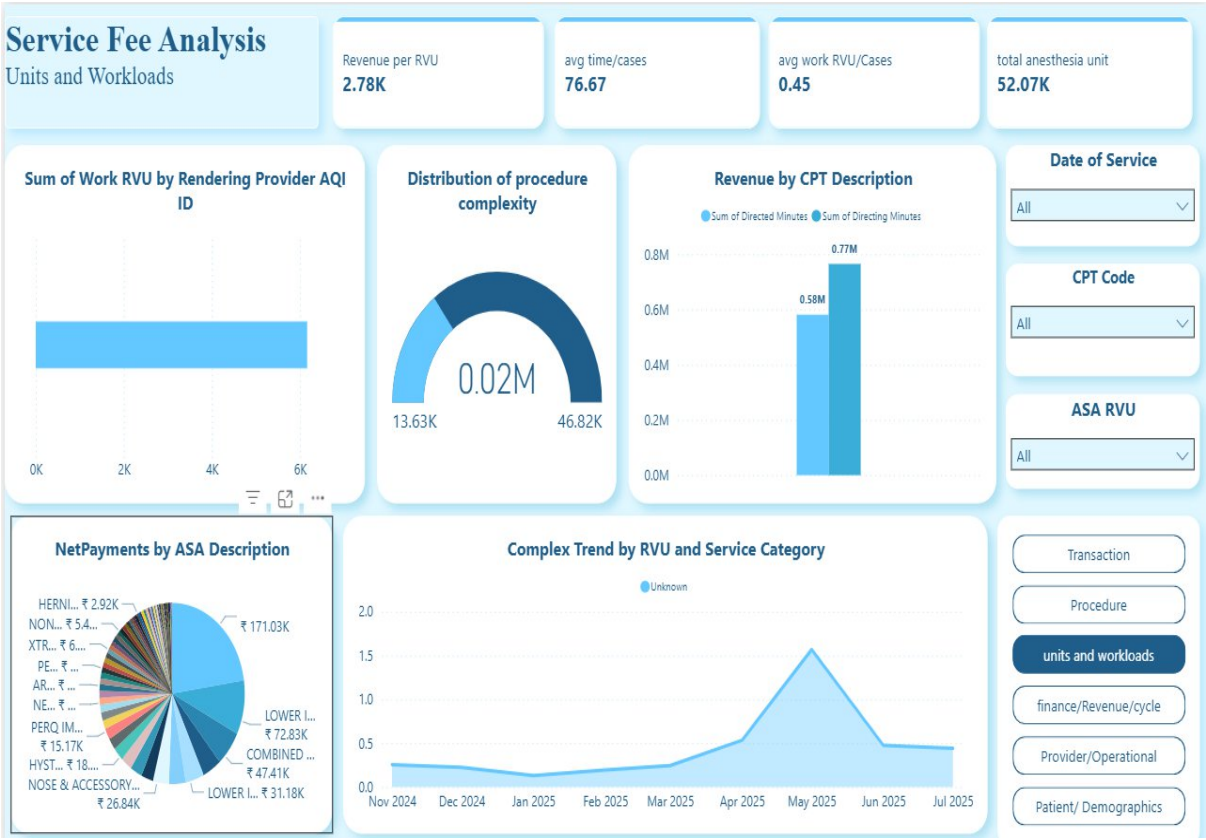

Fig . 03 UNITS AND WORKLOADS

💰 4. Finance Dashboard

🎯 Purpose

To evaluate cash flow, payment cycles, and overall financial health.

📊 Key Visuals

- **Risk of Delayed Cash Flow:** Stacked bar chart showing delays by payer or time.
- **Revenue Realization:** Gauge chart to measure revenue realization against targets.
- **Expected vs Actual Revenue by MRN:** Line chart visualizing collection efficiency.
- **Insurance Revenue Contribution:** Pie chart showing payer-level contribution.
- **Outstanding Payments by Aging Days:** Column chart identifying overdue payments.

📌 KPIs

- Net Payment
- Total Charges
- Outstanding Balance
- Payment per Area Unit

🔍 Slicers

- Aging Days
- MRN
- Primary Payer

💡 Managerial Insights

- Helps monitor payment lags and forecast cash flow.
- Identifies high-risk accounts with large outstanding balances.
- Assists in evaluating insurance performance and improving claim cycles.

4. Finance

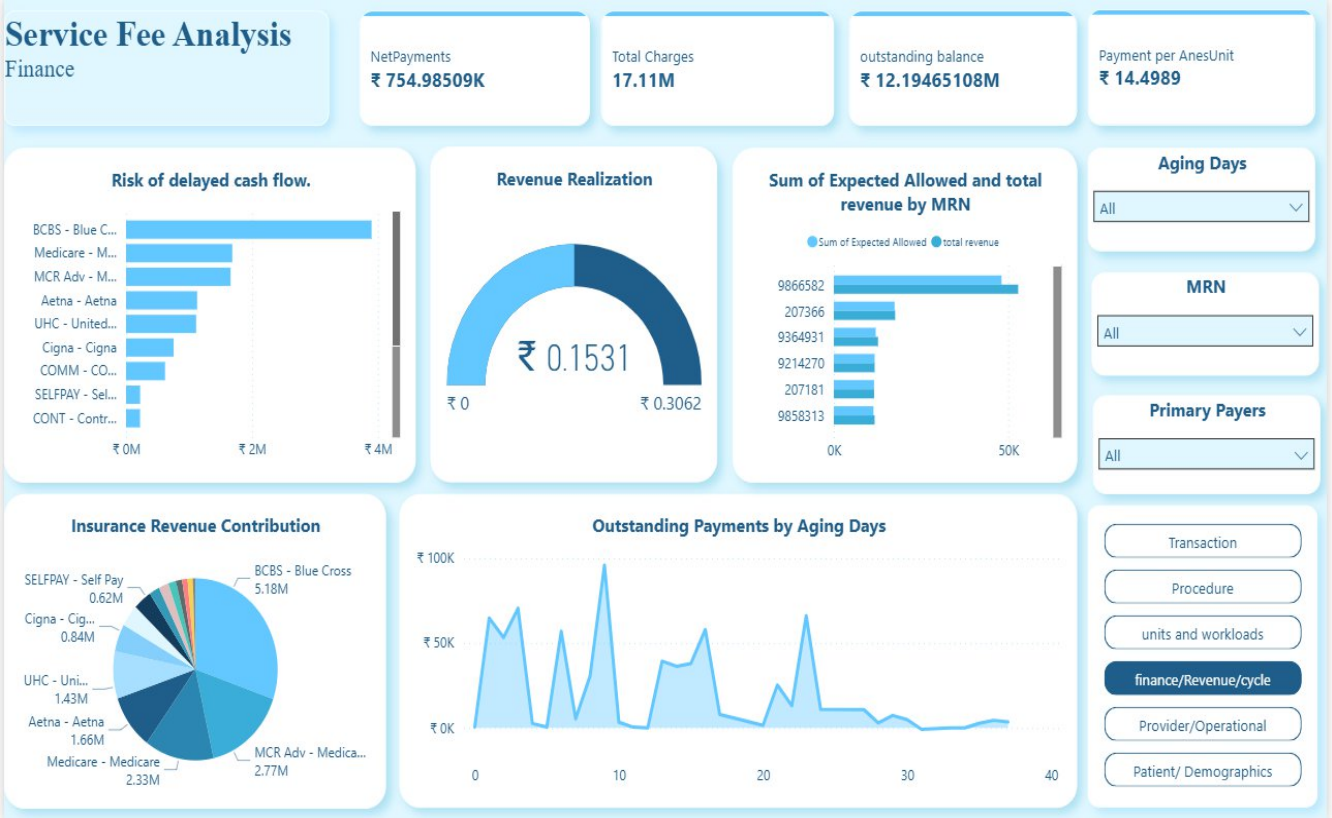

Fig. 04 FINANCE

5. Provider & Operational Dashboard

🎯 Purpose

To assess provider performance, case load, and operational efficiency.

📊 Key Visuals

- **Cases Handled by Provider:** Stacked bar chart indicating workload distribution.
- **Total Units by Provider (Gauge):** Tracks contribution per provider.
- **Direct vs Performed Procedures:** Clustered column chart analyzing service type mix.
- **CPT Code Distribution by Provider:** Pie chart identifying specialization patterns.
- **Peak Load by Time:** Line chart detecting busiest hours/days.

📌 KPIs

- Sum of Processing Days
- Sum of Quantity
- Average Case Duration
- Total Service Rendered

🔍 Slicers

- Date of Service
- CPT Code
- MRN

💡 Managerial Insights

- Highlights top-performing providers by volume and efficiency.
- Optimizes scheduling and workload distribution.
- Supports staffing and resource planning during peak periods.

5. Provider /Operational

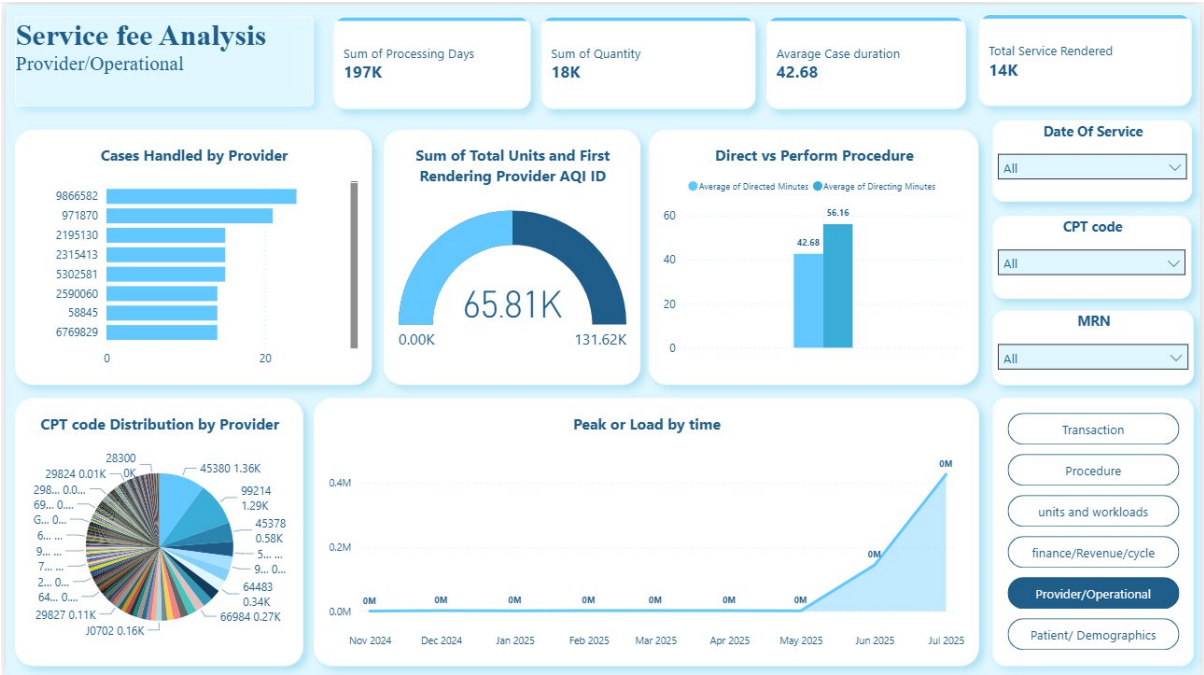

Fig. 05 PROVIDER / OPERATIONAL

□♀ 6. Patient & Demographics Dashboard

🎯 Purpose

To analyze patient characteristics, payment types, and service demand patterns.

📊 Key Visuals

- **Insured vs Self-Pay Comparison:** Clustered column chart showing payer type split.
- **Gender Distribution:** Gauge chart for demographic balance.
- **Patient Volume by Place of Service:** Stacked bar chart mapping service locations.
- **Procedure Planning & Staffing:** Pie chart supporting capacity planning.
- **Seasonal Demand:** Line chart tracking procedure trends over time.

📌 KPIs

- Female Patients
- Male Patients
- Average Work RVU per Patient
- Total Cases

🔍 Slicers

- Date of Service
- Gender
- Type of Service

📢 Managerial Insights

- Enables gender-based and service-type segmentation.
- Supports forecasting for elective or seasonal demand.
- Helps plan anesthesia staffing and optimize patient flow.

7.Patient / Demographic

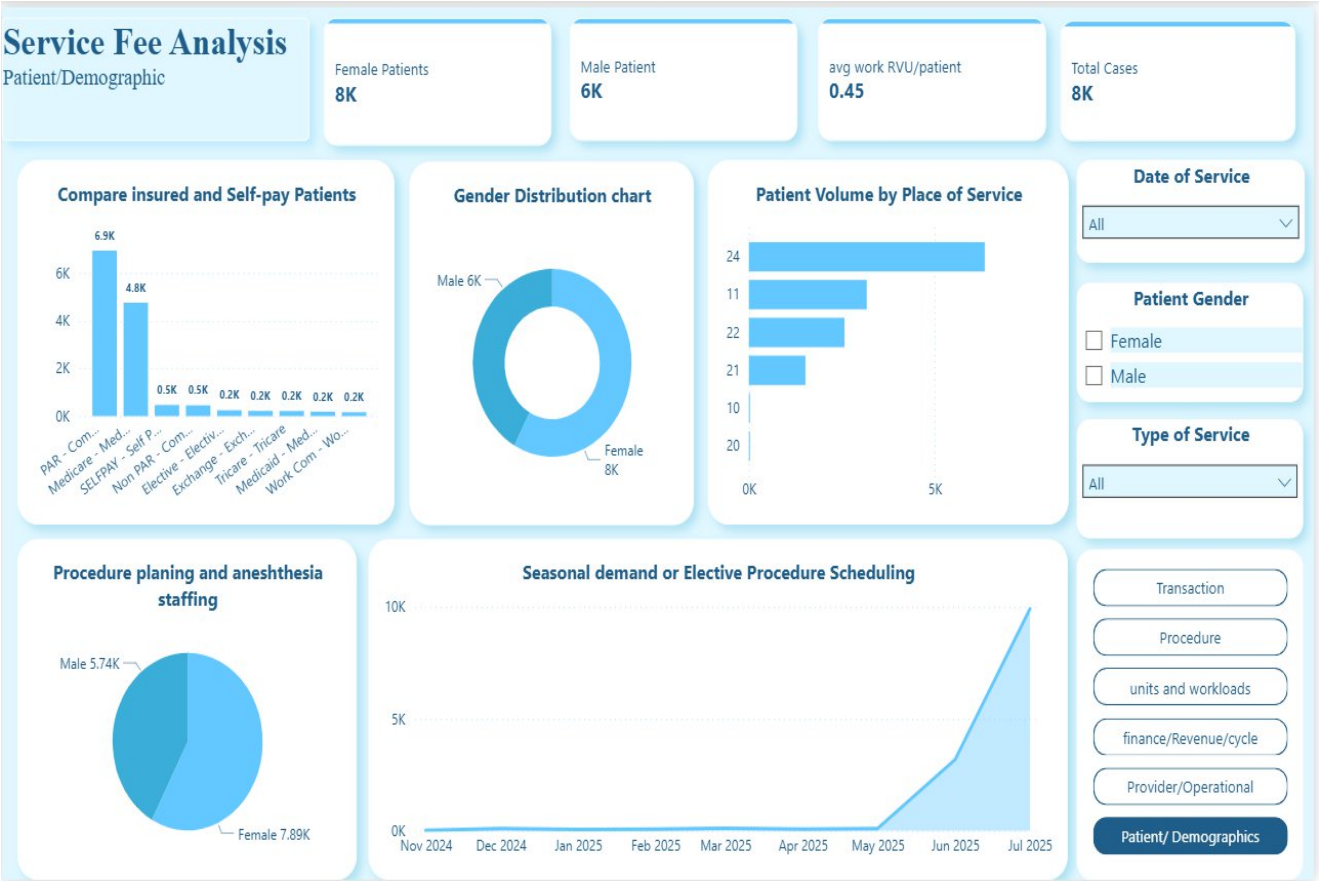

Fig. 06 PATIENT AND DEMOGRAPHICS

7. 🧠 Overall Insights & Business Impact

- **Financial Performance:** Clear identification of top revenue-generating payers and procedures improves strategic billing decisions.
- **Operational Efficiency:** Average processing days, workload trends, and provider-level metrics optimize workforce management.
- **Revenue Assurance:** Cash flow analysis and outstanding tracking strengthen financial stability.
- **Patient-Centric Planning:** Demographic insights assist in service design and staffing alignment.
- **Data-Driven Governance:** Unified dashboards allow leadership to monitor performance across financial, clinical, and operational domains in real time.

📈 Conclusion

The **Service Fee Analysis Report** integrates financial, procedural, operational, and demographic data to create a **360° view** of organizational performance.
With these interactive dashboards and KPIs, managers can:

- Identify inefficiencies,
- Enhance revenue recovery,
- Improve resource allocation, and
- Support strategic decision-making.

This report provides a **robust analytical foundation** for continuous performance monitoring and future predictive analytics implementation.